

Tech Proposal on Strengthen Ticket Security and Prevent Fake Ticket Claims

SA Proposal

Prepared by: Ronald Thian

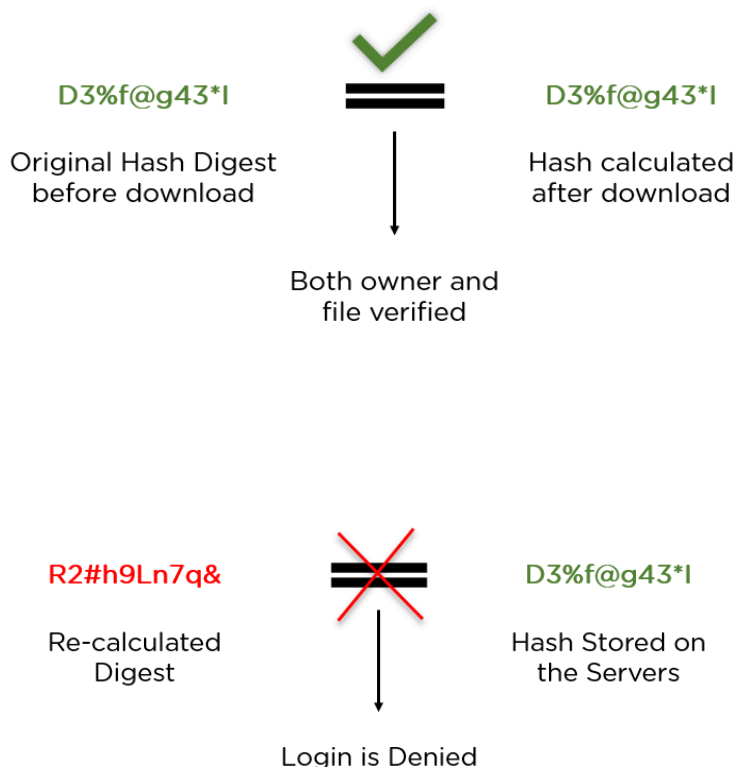
Prepared on: 21 April 2025

Verified with: Karwai (Architecture), Seng Kai (Developer POV), Jazz (Developer POV)

Context

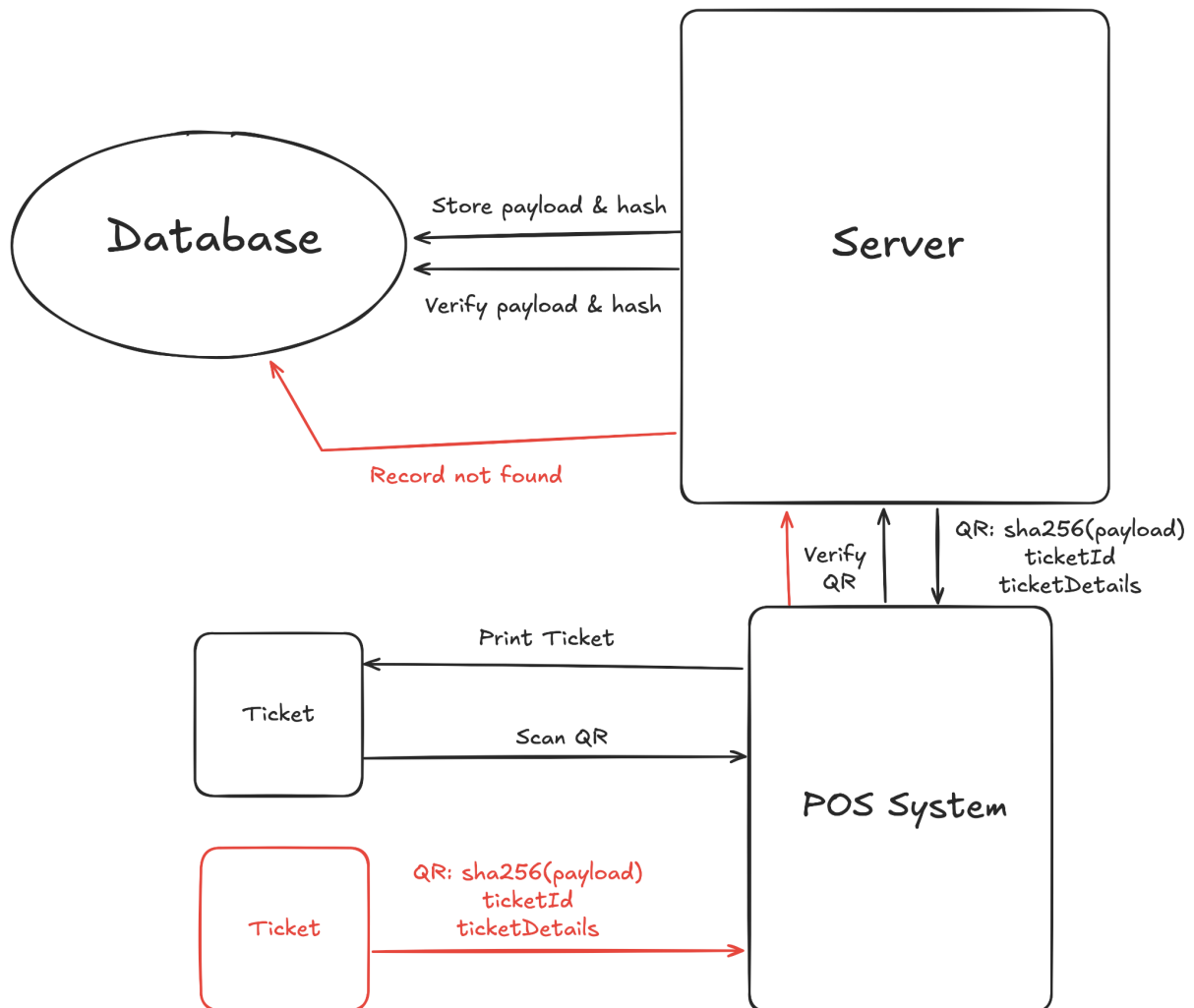
This proposal will mainly focus on the proposed enhancement Item #2: Digital Signature.

As of writing, lottery ticket QR Codes are hashed with SHA-256, a cryptographic hash function most commonly used to ensure data integrity. In a common use case, **two hashes are produced and compared**. One generated by the sender and sent together with the file, and one calculated by the recipient after receiving the file.



This ensures that the file is not manipulated or changed by a third party.

Current Implementation

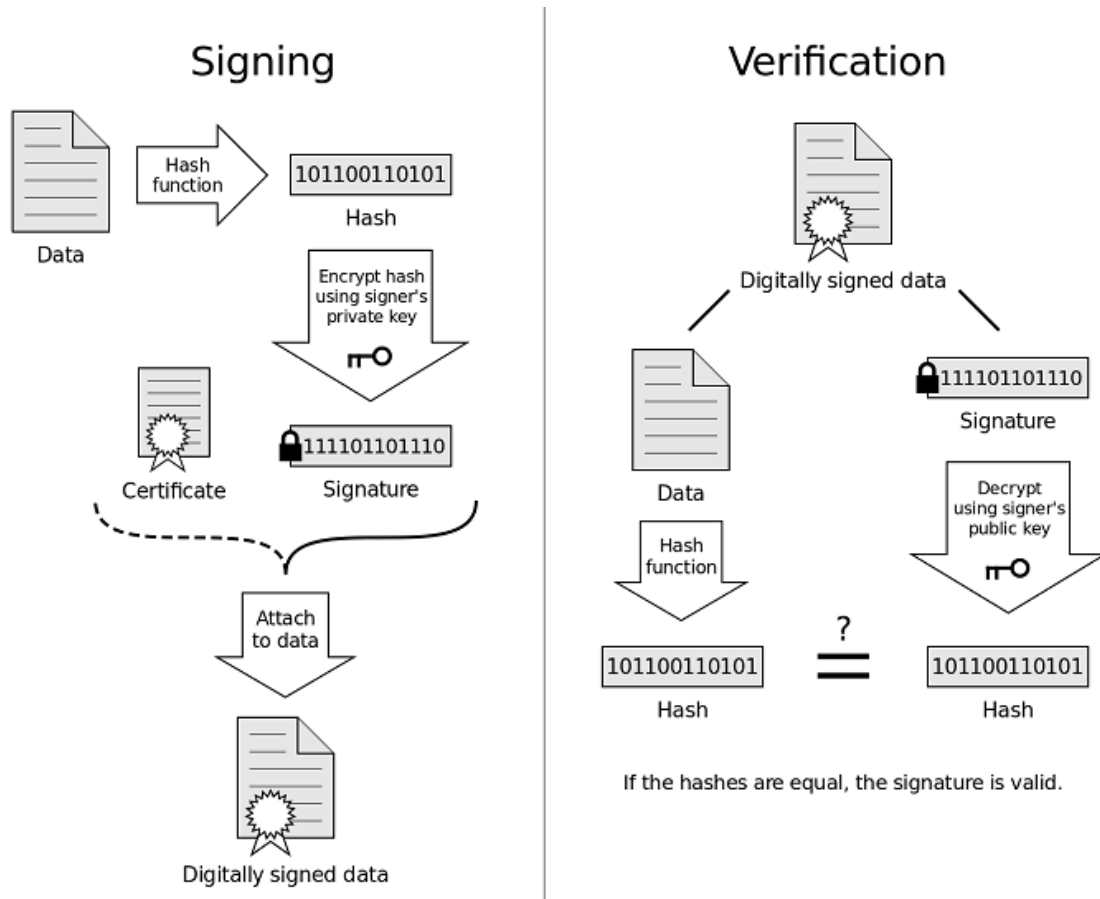


Today, our system uses SHA-256 as an obfuscation method, to hide valuable information from the user, to make it harder to replicate fake tickets. However, the system is unable to confidently identify fake tickets as SHA-256 is an open algorithm, which means it would be challenging **to prove whether a ticket is maliciously faked and printed**, or that there is an underlying system mishap. The only verification layer is at the querying of the database using the SHA-256 hash scanned from the POS System.

Proposal

Context

By leveraging on point #2, we can introduce Digital Signature into our ticketing system, which is able to ensure data integrity of a document.



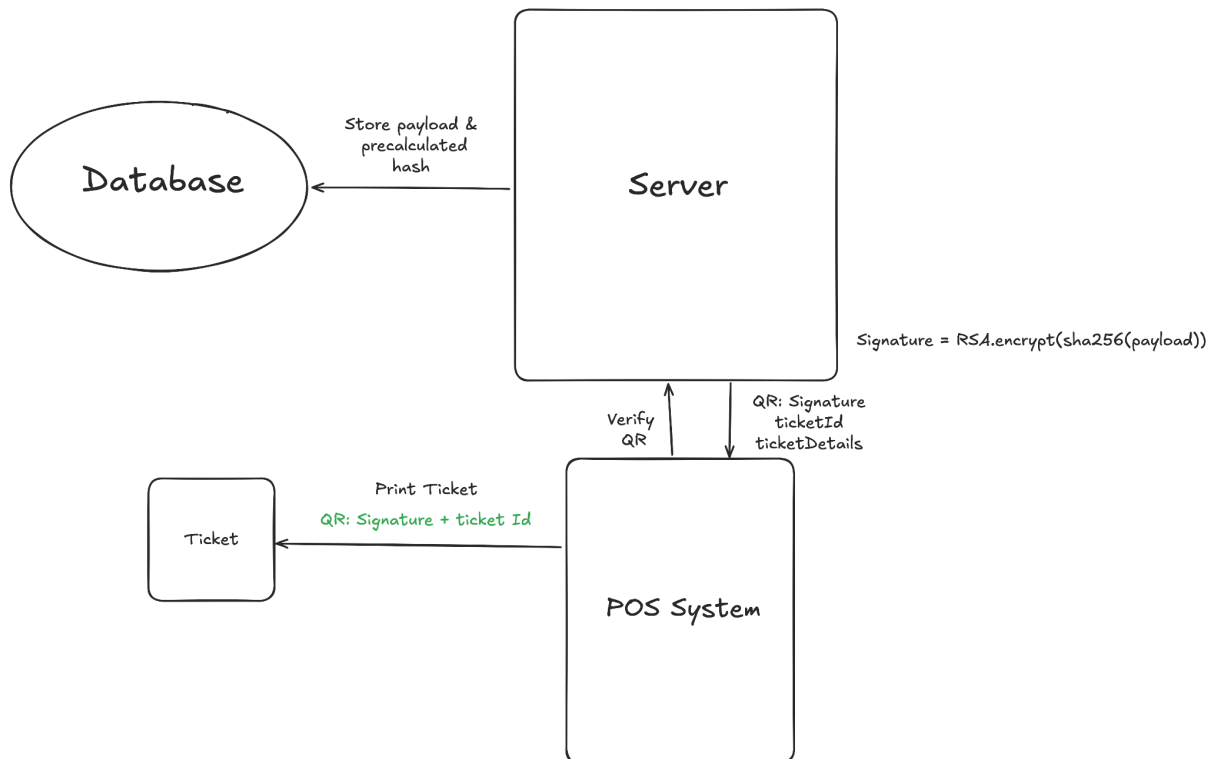
In order to achieve this, the system is required to first calculate the hash of a payload (i.e. Data), and encrypt this hash with an asymmetric private key, producing the Signature. This signature is then attached together with the payload and shipped to the recipient.

During verification, the recipient is able to decrypt the Signature with a public key to obtain a hash. The recipient then generates their own hash of the Data, and by comparing these 2 hashes, the recipient is able to confidently tell if the Data has been manipulated.

Solution

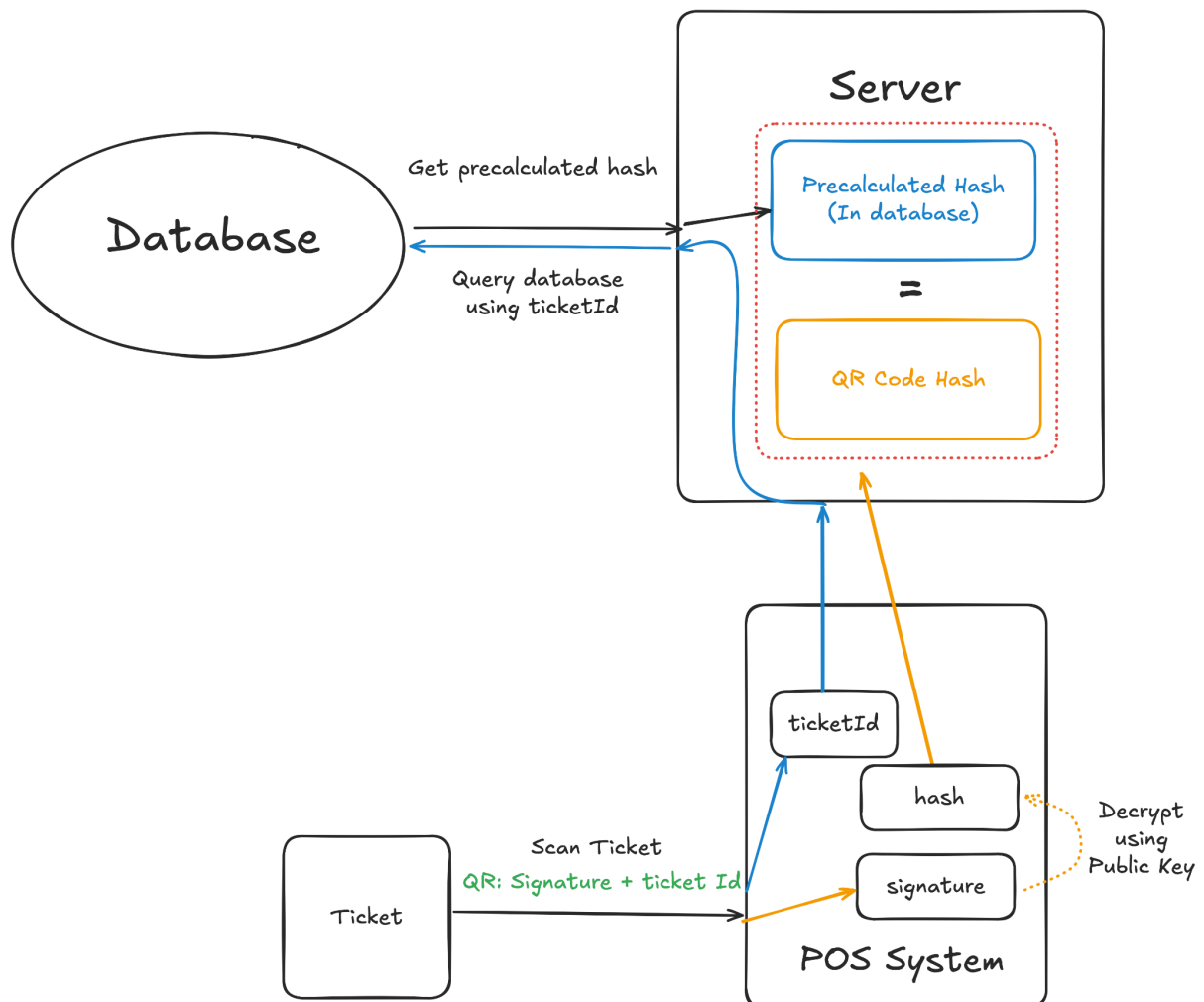
We need to change 2 parts in order to reap the benefits of a digital signature.

1. Firstly, during the point of ticket sale and printing:



- a. Instead of returning the hash directly to be converted into a QR Code, the server should return the RSA Encrypted result, encrypted using a private key stored on the server (i.e. **Signature**).
- b. The POS System should print this encrypted result, with the **ticketId** delimited by an identifiable symbol, allowing itself to quickly extract this 2 information quickly later in the claiming process. (e.g. **"Signature|ticketId"**)

2. For the claiming process, the following changes have to be made:



- By appending the **ticketId** in the QR Code in (1b), we can quickly obtain both the **Signature** and the **ticketId** during scanning.
- Before passing to the server, the POS System is able to decrypt the **Signature** into the file's **hash**
- The POS System then passes both the **hash** and **ticketId** to the server
- The server will then query the Database using the **ticketId** (instead of the current implementation of using **hash**), and grab the precalculated **hash** from that row
- With two hashes, one precalculated and stored in the database, another from the QR Code, the server can confidently perform an authenticity check.

If the check in (2e) fails, the system can confidently say that this **ticket is faked** and not a system mishap, as only the server is able to encrypt and produce a correct **Signature**.

Thoughts on Other Proposed Enhancement

Proposed Enhancement Item #3: Tie Tickets to Transaction Context - Our system is already using our own method to calculate hashes using unique context like `ticketId`, and ticket creation timestamp. This has been acting as another layer to prevent malicious actors from mimicking our ticket creation.

Proposed Enhancement Item #4: Strengthen Agent-Level Authentication - This is less of a concern as the current system already authenticates Agent accounts and Devices, but can be further explored. Here's a comprehensive list of the initiatives and its statuses:

1. Whitelisting IP on POS Devices - Pending DFNN
2. Unique Field during API Calls - Already Implemented, possibility to expand further
3. Agent Account Token Authentication - Already Implemented

Proposed Enhancement Item #5: Out-of-Band Claim Token - Excellent solution provided there is room to compromise on the ticket buying process, as it introduces additional steps like verifying buyers via an alternative method. Here's a list of the possibilities that we may explore:

1. Additional Claim Token sent via SMS/In-app notification
2. Phototaking during ticket point of sales, to tie buyer face with ticket
3. Mandatory filling in of phone numbers to allow system to implement 2FA during claiming process